

Dear Esri Hiring Team,

I am excited to apply for the C++ Software Engineer I – Maps SDKs position at Esri. I am pursuing an M.S. in Applied Machine Learning at the University of Maryland, graduating in May 2027, and have close to two years of software engineering experience at ICBC USA, Bank of China, and ModeSens. The opportunity to develop modern C++ components for a cross-platform mapping API strongly aligns with my systems experience and interest in location-aware applications.

One project that first developed this interest was a cross-platform augmented-reality mobile application that I built for iOS and Android. The application combined GPS-based mapping, real-time Redis-backed data, notifications, and device sensor inputs to create an interactive location-based experience. Building it showed me how strongly the design, performance, and reliability of an underlying mapping API affect the applications and user experiences built on top of it. Esri's work in field mapping, navigation, data collection, and sensor networks therefore feels especially meaningful to me.

I have continued strengthening my modern C++ skills through Quant.ai, a self-built research and execution platform. Its systems core includes a deterministic C++17 event-processing engine, object-oriented limit order book, sequence validation, and TCP/UDP networking with POSIX sockets. I designed the system around clear interfaces, predictable state transitions, and testable invariants. I also used Linux development tools, sanitizers, profiling, and correctness testing to identify performance and memory issues. The engine processed 2.5 million events at more than 507,000 events per second while passing all determinism and invariant checks.

My professional experience has taught me how to build software that remains reliable beyond its initial implementation. At ModeSens, I developed Python and AWS pipelines that integrated product, pricing, and inventory data from more than 30 retailers. At ICBC USA and Bank of China, I worked on financial workflow and transaction systems involving APIs, validation, controlled state transitions, monitoring, and production issue investigation. These experiences strengthened my ability to understand user requirements, communicate complex problems, and improve software based on real operational behavior.

Having grown up and studied across China, Japan, and the United States, I am comfortable adapting to different perspectives and collaborating with people from varied backgrounds. I would be excited to bring my C++ experience, systems mindset, and interest in location technology to Esri while learning more about GIS, cross-platform SDK design, and native application development.

Thank you for your consideration.

Sincerely,

Yuliang Peng